Short Research Article

Exploring the Boundaries of the Number–Temporal Order Association

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Abstract. Past research has shown that numbers are associated with order in time such that performance in a numerical comparison task is enhanced when number pairs appear in ascending order, when the larger number follows the smaller one. This was found in the past for the integers 1–9 (Ben-Meir, Ganor-Stern, & Tzelgov, 2013; Müller & Schwarz, 2008). In the present study we explored whether the advantage for processing numbers in ascending order exists also for fractions and negative numbers. The results demonstrate this advantage for fraction pairs and for integer-fraction pairs. However, the opposite advantage for descending order was found for negative numbers and for positive-negative number pairs. These findings are interpreted in the context of embodied cognition approaches and current theories on the mental representation of fractions and negative numbers.

Keywords: temporal order, numerical comparisons, embodied cognition, numerical representation, negative numbers, fractions

The idea of a tight link between the dimensions of space, time, and number has a long tradition. From a philosophical point of view, Immanuel Kant argued that space, time, and number provide “a priori intuitions” that precede and structure how human beings experience the environment. From a scientific point of view, it has been suggested that the processing of these three dimensions bears strong resemblance to each other, and is conducted by similar brain mechanisms, located in the parietal lobe (e.g., Burr, Ross, Binda, & Morrone, 2010; Dehaene & Brannon, 2010; Walsh, 2003).

A prominent line of research has demonstrated the interaction between numbers and space using the SNARC (Spatial-Numerical Association of Response Codes) effect, which is the finding that small numbers are responded to faster with the left hand, while large numbers are responded to faster with the right hand (Dehaene, Bossini, & Giraux, 1993). It is usually interpreted as evidence for a spatial mental representation of numbers along a number line with small numbers or magnitudes represented on the left and large numbers or magnitudes on the right (e.g., Dehaene et al., 1993; Hubbard, Piazza, Pinel, & Dehaene, 2005; Wood, Willmes, Nuerk, & Fischer, 2008).

There is also evidence for an association between space and time. It has been shown that the representation of the past is associated with the left side, and the represented future is associated with the right side (Bonato & Umilta, 2014; Bonato, Zorzi, & Umiltà, 2012; Oliveri et al., 2009; Ouellet, Santiago, Funes, & Lupiáñez, 2010; Ouellet, Santiago, Israeli, & Gabay, 2010; Santiago, Lupáñez, Pérez, & Funes, 2007). In a similar manner, shorter durations were found to be associated with the left side, while longer durations with the right side (Vallesi, Binns, & Shallice, 2008).

Research has also provided evidence for the last part of this triangle, which is the number-time association (see Bonato et al., 2012 for a review). When judging the duration of events, participants are consistently affected by irrelevant magnitude information, with smaller magnitudes associated with shorter durations (Cappelletti, Freeman, & Cipolotti, 2009; Dormal, Seron, & Pesenti, 2006; Droit-Volet, Clement, & Fayol, 2003; Oliveri et al., 2008; Vicario, 2011; Xuan, Zhang, He, & Chen, 2007).

The Number–Temporal Order Association

More relevant for the present work are studies that examined the association between magnitude and temporal order. Such studies have shown enhanced performance in a numerical comparison task when the numbers are presented in ascending versus descending order (Ben-Meir et al., 2013; Kaan, 2005; Müller & Schwarz, 2008). This pattern is termed here *the ascending order advantage*. It suggests the mapping of magnitude into temporal order, such that “small” is associated with “early,” and “large” with “late.”

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The ascending order advantage is not limited to numbers as it was found also when participants judged physical magnitude (Ben-Meir et al., 2013).

So far the ascending order advantage for numerical magnitudes was mainly found for integers from the first decade. The goal of the present study was to explore whether it could be found for numbers beyond the first decade. Specifically, we examined whether it exists for fractions and for negative numbers. Before proceeding into describing the present study and its predictions, a brief summary of recent results on fractions and negative numbers is provided.

Processing Fractions and Negative Numbers

Past research examined whether fractions as well as negative numbers are represented holistically or in terms of their components. A related question was whether they are represented on the same number line as integers. There seems to be enough evidence to conclude that fractions are generally represented holistically (Ganor-Stern, Karasik-Rivkin, & Tzelgov, 2011; Meert, Gregoire, & Noël, 2010; Schneider & Siegler, 2010), on the same mental number line with integers (Ganor-Stern, 2012, 2013; Kallai & Tzelgov, 2009). This is indicated by the presence of distance effects in both fraction comparisons and integer-fraction comparisons (but see also Bonato, Fabbri, Umilta, & Zorzi, 2007).

The picture for negative numbers is less clear (Blair, Rosenberg-Lee, Tsang, Schwartz, & Menon, 2012; Gullick, Wolford, & Temple, 2012; Kong, Zhao, & You, 2012; Varma & Schwartz, 2011). Some studies found evidence for a components representation (Ganor-Stern & Tzelgov, 2008; Kong et al., 2012; Tzelgov, Ganor-Stern, & Maymon-Schreiber, 2009), while others support the holistic representation (Fischer, 2003; Shaki & Petrusic, 2005). Importantly, there is more evidence for a holistic representation of negative numbers when the numbers are presented sequentially, and thus both components of the negative number (the sign and the digit) had to be held in memory (Ganor-Stern, Pinhas, Kallai, & Tzelgov, 2010).

It is suggested that the issues of the numerical representations of fractions and negative numbers and the ascending order advantage are linked. The ascending order advantage will be present for negative numbers and fractions only if they are represented holistically. Furthermore, the same effect for mixed pairs (composed of an integer and a fraction/negative number) will be present only if they are represented holistically on the same number line. Past research has shown that presenting numbers sequentially (as done in the present study) encourages the formation of a holistic representation (Ganor-Stern et al., 2010; Ganor-Stern, Pinhas, & Tzelgov, 2009). Note, however, that a holistic representation is a necessary condition for the ascending order advantage, but not a sufficient one. Numbers might be represented holistically, but not mapped into temporal order. The mapping of number into temporal order might be limited to some number types.

The Ascending Order Advantage for Fractions and Negative Numbers

Exploring the ascending order advantage for negative numbers and fractions will provide information about the generality of this effect and it will also help to disentangle two possible explanations for the origin of the effect. The first is the *forward association* hypothesis. According to this account, the advantage for numbers in ascending order is due to the lexical associations between consecutive numbers (Müller & Schwarz, 2008). Each integer provides forward activation to the next numbers in the counting sequence, and thus facilitates the processing of numbers in ascending order.

For unit fractions and negative numbers there is a negative relationship between the magnitude of the whole number and that of its components. (e.g., 1/3 is larger than 1/7, and −3 is larger than −7, although 7 is larger than 3). Thus, if the associative links between the digits account for the ascending order advantage for integers then it should be either absent or reversed for negative number comparisons and fraction comparisons. As there is no reason to assume associative links between positive and negative numbers and between fractions and integers, the advantage for ascending order should be absent for mixed pairs.

The second explanation focuses on the representation of magnitude and its association to the representations of time and space (Walsh, 2003). The ascending order advantage might be grounded in the interaction with the world. People learn from observation that things, especially living things, tend to grow larger with time. Support for this idea is provided by the presence of the ascending order advantage not only for numerical magnitude but also for physical magnitude (Ben-Meir et al., 2013). This explanation is in line with embodied cognition approaches, which argue that human cognition, rather than being centralized, abstract, and sharply distinct from peripheral input and output modules, may have deep roots in sensorimotor processing (Lakoff & Nunez, 2000; Wilson, 2002).

According to this explanation, the association between magnitude and temporal order, which is rooted in the physical world, is generalized to numbers that are symbols used to represent quantities or magnitudes. Two accounts might be proposed with respect to the generalization of this association. According to the *quantity account* it is generalized only to numbers representing quantities that can be perceived in the real world, such as integers and fractions, and thus, the ascending order advantage will be found only for such numbers. For numbers that do not represent quantities (i.e., negative number pairs), there should be no such advantage, or there should even be a reversed advantage for numbers in descending order, which is in ascending order of the negative number components. Alternatively, according to the *magnitude account* – the mapping of number to temporal order might be generalized to all numbers, and it might be part of any numerical representation. If so, the ascending order advantage will be found for *all* numbers that are represented holistically, even for negative numbers that represent magnitudes that cannot be perceived in the real world (Ganor-Stern, 2012; Siegler & Lortie-Forgues, 2014).

To summarize, the forward association, the quantity, and the magnitude accounts would all predict the ascending order advantage for integers. The forward association explanation would predict that the ascending order advantage will be present *only* for integer pairs. The magnitude account would predict that it will be present for *all* pair types, while the quantity account would predict that it will be present for integer pairs, fraction pairs, and integer-fraction pairs, but not for negative number pairs or positive-negative mixed pairs.

In the present study, pairs of numbers were presented sequentially for a numerical comparison task. Experiment 1 included pairs of integers, pairs of fractions, and mixed fraction-integer pairs. Experiment 2 included pairs of positive numbers, pairs of negative numbers, and mixed positive-negative number pairs. The number set included the integers 1–9 in both experiments, the unit fractions 1/9–1/2 in Experiment 1, and the negative numbers -9 to -1 in Experiment 2. Only unit fractions were used in Experiment 1 so that the same components were used in both Experiments 1 and 2. To explore effects of instructions, both *select larger* and *select smaller* instructions were used. To investigate possible modulation of the magnitude-temporal order association by space (Bonato et al., 2012), two response mappings were used (see Methods Section for details).

Experiment 1

Method

Participants

Forty Psychology students (32 females, mean age 23.2 years) participated in Experiment 1 for course credit. All were native Hebrew speakers, with normal or corrected-to-normal vision and without learning disabilities.

Stimuli

The numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, 1/2, 1/3, 1/4, 1/5, 1/6, 1/7, 1/8, and 1/9 were used. The numbers were printed in white on a black background. The integers were about 40 mm high and 20–25 mm wide. Each unit fraction was 80 mm high and 25 mm wide. The size of each fraction component was 30 mm high and 20–25 mm wide. The distance between the centers of the numbers within each pair was 110 mm. The set of stimuli was composed of all 136 possible unique pairs of the different numbers. These pairs were divided into three pair types – 36 integer pairs, 28 fraction pairs, and 72 mixed pairs composed of an integer and a fraction.

Procedure

Participants performed the select larger and the select smaller instructions in separate blocks. The order of blocks was counterbalanced across participants. The A and L keys on a QWERTY keyboard were used as response keys. Half of the participants were instructed to press the left (i.e., "A") key, if the first number was larger (smaller) than the second number and the right (i.e., "L") key, if the second number was larger (smaller) than the first number. The other half were asked to press the right (i.e., "L") key, if the first number was larger (smaller) than the second number and the left (i.e., "A") key, if the second number was larger (smaller) than the first number. Participants were required to respond as fast as possible but to avoid errors.

Each task began with six practice trials, followed by the experimental trials in random order. In each trial a fixation cross appeared at the center of the screen for 300 ms, followed by the first digit that appeared for 500 ms, followed by a blank screen for 300 ms. Then the second digit appeared and stayed in view until the participant's response. The second digit appeared in the same location as the first one. The next trial began 400 ms after participant's response. In each block, each pair appeared twice, once in ascending order and once in descending order. Thus, there were 272 trials in each block and 544 trials in the whole experiment.

Apparatus

This and the following experiment were programmed with E-Prime software (Schneider, Eschman, & Zuccolotto, 2002) and run on IBM clone, Pentium III computers with 17-inch monitors.

Results and Discussion

Means of reaction times (RTs) of correct responses and error rates (ERs) were analyzed separately. Significance level was defined as $p < .05$. Partial eta squared values (η_p^2) were calculated as SSeffect/(SSeffect + SSerror). Mean RTs and ERs were submitted to a three-way analysis of variance (ANOVA) with temporal order of numerical size (ascending vs. descending), pair type (integers, fractions, mixed integer-fraction), and instructions (select smaller vs. select larger), as within-participants variables and mapping of responses (mapping 1: If second number is larger/smaller than first number press the left key A, if first number is larger/smaller than second number press the right key L; mapping 2: If second number is larger/smaller than first number press the right key L, if first number is larger/smaller than second number press the left key A) as a between-participants variable (see Electronic Supplementary Materials, ESM 1 and 2).

Table 1. Mean Response Times (RTs) and Error Rates (ERs) by pair type and temporal order in Experiment 1

	RT		ER	
Pair type	Ascending	Descending	Ascending	Descending
Integer	747	758	0.06	0.08
Fraction	958	1,001	0.10	0.14
Mixed integer-fraction	730	738	0.04	0.06

Note. Response time is measured in milliseconds.

Trials with RTs longer than 3,000 ms were excluded from the analyses (less than 0.3% of the trials). Moreover, trials with the first digit being 9, 8, 1/9, or 1/8 were also excluded from the analyses because the response could be made with certainty or with very high probability prior to the presentation of the second number.

RT Analysis

There was a main effect of pair type, $F_{(2,76)} = 103.48$, $MSE = 28{,}843$, $\eta_p^2 = .73$. Mixed pairs were responded to faster than integer pairs and fraction pairs, $F_{(1,38)} = 133.40$, $MSE = 13{,}978$, $\eta_p^2 = .77$. Integer pairs were responded to faster than fraction pairs, $F_{(1,38)} = 93.91$, $MSE = 43{,}703$, $\eta_p^2 = .71$. Importantly, there was a main effect of temporal order, $F_{(1,38)} = 5.36$, $MSE = 9{,}537$, $\eta_p^2 = .12$, as comparisons were faster when the numbers appeared in ascending versus descending order (Table 1).

ER Analysis

ER varied by pair type, $F_{(2,76)} = 29.07$, $MSE = 0.006$, $\eta_p^2 = .43$. ER was smaller for mixed pairs than for integer pairs and fraction pairs, $F_{(1,38)} = 27.24$, $MSE = 0.006$, $\eta_p^2 = .42$. ER was higher for fraction pairs compared to integer pairs, $F_{(1,38)} = 30.96$, $MSE = 0.006$, $\eta_p^2 = .45$. More importantly, ER was lower for number pairs presented in ascending (0.07) versus descending order (0.09), $F_{(1,38)} = 9.12$, $MSE = 0.008$, $\eta_p^2 = .19$.

There was also a significant interaction between instructions, mapping, and temporal order in ER, $F_{(2,76)} = 4.24$, $MSE = 0.005$, $\eta_p^2 = .10$, indicating the fact that the ascending order advantage was present with the select larger instruction under both mappings, but it was present with select smaller instructions only when participants had to press with the left hand for the smaller number. There was also a significant interaction between instructions, pair type, and mapping, $F_{(2,76)} = 3.75$, $MSE = 0.005$, $\eta_p^2 = .09$. However, since it did not include the temporal order variable, it was not analyzed further.

The main finding of Experiment 1 was the presence of the ascending order advantage for all pair types – integer pairs, fraction pairs, and mixed pairs. This advantage was reflected in both response latency and error rate. It expands on past research that showed the ascending order advantage only for integers (Ben-Meir et al., 2013; Müller & Schwarz, 2008).

The presence of the ascending order advantage for fraction pairs is not obvious because for unit fraction pairs, ascending order of the fractions means descending order of the fraction components. Thus, for the ascending order advantage to be found for fraction pairs, the holistic representation should be stronger than the components one (Schneider & Siegler, 2010). Furthermore, for the effect to be found for integer-fraction pairs, fractions should be represented holistically on the same number line with integers (Ganor-Stern, 2012; Kallai & Tzelgov, 2009).

The fact that the ascending order advantage was found for both unit fraction pairs and mixed pairs goes against the forward association account and the idea that this advantage is due to associative links between the numbers, as there is no reason to assume associative links between a fraction and an integer, and between the unit fractions.

The alternative explanation proposed for the ascending order advantage is that it is based on our everyday functioning in the world (Lakoff & Nunez, 2000), which shows us that things tend to grow larger with time. This knowledge is generalized to all numbers according to the magnitude hypothesis, or only to numbers that represent quantity according to the quantity hypothesis. According to the quantity account, for numbers that do not represent quantities, such as negative numbers, the mapping to temporal order might be based on the numbers' components and not on the whole numbers, and thus there should be an advantage for numbers in descending order. These predictions were tested in Experiment 2, which included negative number pairs, positive number pairs, and mixed positive-negative number pairs.

Experiment 2

Method

Participants

Thirty-two new participants (20 females, mean age 23.1 years) from the same participant pool as Experiment 1 took part in Experiment 2.

Stimuli

The numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, −1, −2, −3, −4, −5, −6, −7, −8, and −9 were used. The numbers were printed in white on a black background. The integers were about 40 mm high and 20–25 mm wide. The polarity marker of the negative number was 5 mm high and 15 mm wide. The distance between the centers of the numbers within each pair was 110 mm. The set of stimuli was composed of all 153 unique possible pairs of the different numbers. These pairs were divided into three pair types – 36 positive number pairs, 36 negative number pairs, and 81 mixed pairs composed of one positive and one negative number.

Procedure and Apparatus

The procedure and apparatus were identical to the ones used in Experiment 1, except for the fact that there were 306 trials in each block and 612 trials in the whole experiment.

Results and Discussion

Means of RTs of correct responses and ERs were analyzed in the same way as in Experiment 1 with temporal order of numerical size, pair type, and instructions as within-participants variables and mapping of responses as a between-participants variable. Trials with RT longer than 3,000 ms were excluded from the analyses (less than 0.5% of the trials). Moreover, trials with the first digit being 9, 8, −9, or −8 were excluded from the analyses because in such cases the response could be made with certainty or with very high probability prior to the presentation of the second number. The results are presented in Table 2 (see also Electronic Supplementary Material, ESM 3).

RT Analysis

There was a main effect of pair type, $F_{(2,60)} = 47.70$, $MSE = 13{,}791$, $\eta_p^2 = .61$. Mixed pairs were responded to faster than pairs of positive numbers and pairs of negative numbers, $F_{(1,30)} = 115.67$, $MSE = 5{,}956$, $\eta_p^2 = .79$. Positive numbers were responded to faster than negative numbers, $F_{(1,30)} = 28.98$, $MSE = 21{,}625$, $\eta_p^2 = .49$. The effect of pair type interacted with that of instructions, $F_{(2,60)} = 25.37$, $MSE = 6{,}391$, $\eta_p^2 = .49$, such that responses for negative number pairs were faster with the instructions select smaller versus select larger, $F_{(1,30)} = 8.59$, $MSE = 32{,}881$, $\eta_p^2 = .22$. The opposite pattern appeared for positive number pairs although it failed to reach significance. Most important was the interaction between pair type and temporal order, $F_{(2,60)} = 7.37$, $MSE = 9{,}242$, $\eta_p^2 = .20$. It indicated that for positive number pairs, responses were faster for pairs in ascending versus descending order, $F_{(1,30)} = 6.31$, $MSE = 12{,}078$, $\eta_p^2 = .17$, while for negative number pairs and mixed pairs the opposite pattern emerged, with responses faster when the numbers appeared in descending versus ascending order; $F_{(1,30)} = 4.58$, $MSE = 7{,}628$, $\eta_p^2 = .13$, for negative numbers, and $F_{(1,30)} = 5.11$, $MSE = 5{,}302$, $\eta_p^2 = .15$, for mixed pairs. The triple interaction between temporal order, instructions, and pair type, $F_{(2,60)} = 6.89$, $MSE = 4{,}531$, $\eta_p^2 = .19$, indicated that while for positive and mixed pairs the effect of temporal order on RT was similar for both instructions, for negative numbers, the responses were faster for descending versus ascending order mainly with the select smaller instructions, $F_{(1,30)} = 16.12$, $MSE = 7{,}119$, $\eta_p^2 = .35$.

ER Analysis

ER varied by pair type, $F_{(2,60)} = 5.07$, $MSE = 0.005$, $\eta_p^2 = .14$. It was smaller for mixed pairs than for positive number pairs and negative number pairs, $F_{(1,30)} = 14.77$, $MSE = 0.003$, $\eta_p^2 = .33$), which did not differ. There was a significant interaction between temporal order and mapping, $F_{(1,30)} = 4.39$, $MSE = 0.003$, $\eta_p^2 = .13$. It indicated that ER was lower when the numbers were in ascending order in one of the mappings, while the reversed effect was present in the other mapping. However, none of the simple main effects of temporal order were significant. Finally, there was also a triple interaction between pair type, instructions, and mapping, $F_{(1,30)} = 5.07$, $MSE = 0.003$, $\eta_p^2 = .14$, which bore no theoretical importance and thus was not analyzed.

Thus, the main findings of Experiment 2 are the following. First, comparisons of positive numbers were faster when the numbers appeared in *ascending* order. Second, comparisons of negative number pairs or mixed pairs were faster when the numbers appeared in *descending* order.

The first finding on positive number pairs replicated and extended the results of Experiment 1 and past

Table 2. Mean Response Times (RTs) and Error Rates (ERs) by pair type and temporal order in Experiment 2

	RT		ER	
Pair type	Ascending	Descending	Ascending	Descending
Positive numbers	720	769	0.07	0.08
Negative numbers	860	827	0.08	0.07
Mixed positive – negative numbers	719	690	0.05	0.05

Note. Response time is measured in milliseconds.

findings of Müller and Schwarz (2008) and Ben-Meir et al. (2013). The findings on negative number pairs and mixed pairs are novel. Negative number pairs were compared faster when they were presented in descending versus ascending order. Thus, the advantage for ascending order was present for the negative number components but not for its whole value. This is consistent with the quantity account.

An alternative explanation that could be raised is that the advantage for descending order found for negative numbers is due to the fact that the negative numbers were not represented holistically, but only in terms of their components. To rule out this explanation, two indications for a holistic representation of negative numbers are presented. The first is the distance effect in positive-negative number pairs. The presence of such an effect suggests that they were represented on the same continuum and thus implies that negative numbers were represented holistically (Ganor-Stern et al., 2010). An analysis of RTs for such mixed pairs with numerical distance as a within-participants variable revealed a main effect of distance, $F_{(14,434)} = 2.83$, $MSE = 8{,}803$, $\eta_p^2 = .08$. Linear trend analysis showed that RT decreased with increased distance, $F_{(1,31)} = 555.82$, $MSE = 429{,}421$, $\eta_p^2 = .95$. The second indication for a holistic representation is the semantic congruity effect (Banks, Fujii, & Kayra-Stuart, 1976), which is the pattern of faster comparisons of small numbers with the select smaller instructions versus select larger, and faster comparisons of large numbers with the select larger instructions versus select smaller. This pattern was indeed found in this experiment although only the difference for negative number pairs reached significance. The presence of such a pattern for negative numbers suggests that they were perceived as smaller than positive ones, which supports the idea of a holistic representation of both positive and negative numbers on the same number line. Note that this holistic representation most likely exists together with a components representation (Ganor-Stern et al., 2010; Tzelgov et al., 2009). The advantage for negative number pairs presented in *descending* order suggests that the mapping of number to temporal order might not exist for negative numbers, despite their holistic representation. This is consistent with the quantity account, which postulates the existence of the mapping of number to temporal order only for numbers that represent quantity.

As for mixed pairs, they were also compared faster when they were presented in descending versus ascending order. There was no a priori reason for participants to expect positive numbers to be followed by negative ones. The explanation might lie in the components representation that presumably exists side by side with the holistic one. The components representation includes the minus polarity sign separate from the digit. When the first number is positive and the second is negative it is sufficient to process the minus sign of the second number to respond (Ganor-Stern et al., 2010; Tzelgov et al., 2009). When the order is reversed and the first number is negative while the second is positive, there is no polarity sign, and thus participants process the digit. This might take longer than the processing of the minus sign, and it might produce the advantage for descending order in mixed pairs.

General Discussion

The results of the present study are clear cut. The ascending order advantage was found for pairs of integers, pairs of fractions, and mixed integer-fraction pairs. The opposite pattern of an advantage for descending order was found for pairs of negative numbers and for mixed pairs of positive-negative numbers.

Since fractions are not part of the counting sequence, there should be no lexical associations between the fractions and between the fractions and integers. The fact that the ascending order advantage was found for fractions and fraction-integer pairs suggests that lexical forward associations cannot provide the explanation for the ascending order advantage for numbers.

For negative number pairs an advantage was found for numbers in *descending* order despite the fact that the negative numbers were represented holistically. Evidence for this holistic representation is provided by the presence of a distance effect in positive-negative number pairs and by the advantage for select smaller instructions for negative number pairs, consistent with the semantic congruity effect (Banks et al., 1976).

The results are thus consistent with the quantity account, which postulates that the time-magnitude association is limited to numbers that represent quantity. Such an interpretation is in line with embodied cognition approaches according to which cognitive processes take place in the context of a real-world environment, and they inherently involve perception and action (Lakoff & Nunez, 2000; Walsh, 2003; Wilson, 2002). Since negative numbers do not represent quantities that can be perceived by the sensory system, they do not show the temporal order-magnitude association that originates from the experiences with the actual world. In contrast, the digits that are the negative number components do show this association, as reflected in the advantage for descending order found for negative number pairs.

Although negative numbers and fractions share important characteristics, as they are both complex numbers for which the magnitude of the components has a negative relation with the magnitude of the whole number, their association with temporal order differed. A similar proposal of a division between numbers that represent quantities (i.e., integers and fractions) and those that do not (i.e., negative numbers) was made by Ganor-Stern (2012), based on the finding that integer-fraction pairs show a distance effect while positive-negative numbers do not. This finding led to the conclusion that fractions are represented on the mental number line with integers but negative numbers are not. The association between magnitude and temporal order provides a similar picture. Fractions are mapped into temporal order, but negative numbers are not, as indicated by the presence of the ascending order advantage for the former but not for the latter.

Two theoretical frameworks were presented by Bonato et al. (2012) to account for the association between magnitude and time. The first is the theory of magnitude (ATOM) proposed by Walsh (2003), according to which this association reflects the common system for the processing of magnitude, time, and space. An alternative theoretical framework proposed by Bonato and Umilta (2014) postulates the existence of a temporal number line which is spatially represented, with the left side associated with shorter durations or with events that occurred earlier in time, and the right side is associated with longer durations or with events that occurred later in time (Vallesi et al., 2008; Vicario, Caltagirone, & Oliveri, 2007; Vicario et al., 2008). Although these two frameworks share important assumptions, in the present context, the temporal line framework would predict that the association between temporal order and magnitude will be modulated by spatial characteristics of either stimuli or responses, while ATOM would not make a similar prediction. The fact that the ascending order advantage was found in the present study irrespective of response mapping seems to be more consistent with ATOM as the source of the number-temporal order association. Furthermore, the fact that this association was found only for numbers that represent quantity is also in line with ATOM, which postulates the existence of a shared system for encoding information about magnitudes in the external world (i.e., quantity, space, and time) that are used in action.

The present study was conducted on Hebrew speakers that read from right to left. Past studies failed to find a SNARC effect for such participants suggesting that the origin of the association between magnitude and space is at least in part due to reading directions (Shaki, Fischer, & Petrusic, 2009). Note however that for the magnitude-temporal order association the same advantage for ascending order was found for participants who read from left to right and for those who read from right to left (Ben-Meir et al., 2013; Müller & Schwarz, 2008), thus suggesting that the association with time is less affected by cultural factors than the association of magnitude with space.

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Electronic Supplementary Material

The electronic supplementary material is available with the online version of the article at http://dx.doi.org/10.1027/1618-3169/a000285

ESM 1. Data file
This file contains the data submitted to the error analysis of Experiment 1.

ESM 2. Data file
This file contains the data submitted to the RT analysis of Experiment 1.

ESM 3. Data file
This file contains the data submitted to the error and RT analyses of Experiment 2.

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